British Poultry Science Volume 47, Number 3 (June 2006), pp. 330–335

Taylor & Francis
Taylor & Francis Group

End-weight prediction in broiler growth

Ö. CANGAR, J.-M. AERTS, E. VRANKEN AND D. BERCKMANS

Laboratory of Agricultural Buildings Research, Catholic University of Leuven, Leuven, Belgium

Abstract 1. Two different methods, categorised as input–output and single output models, were evaluated for slaughter weight prediction of broiler chickens. The input–output models included linear and non-linear recursive modelling with a time-varying model structure, whereas the output models consisted only of empirical growth equations and several growth curve fitting techniques.
2. The results suggested that a simple linear growth curve fitting method gives the greatest accuracy in a prediction horizon of 4 d or less. Error is minimised to an average of 0·14% when 4 d of past information is used to fit a line to predict the end weight one day ahead.

INTRODUCTION

Accurate prediction of growth patterns and slaughter weight in broiler chickens is essential for broiler producers. Good predictions of end weight, which take into account the weight deviations due to handling and transportation, can improve the efficiency and profitability of the processing plant. By following such models, broiler growers can make better management decisions in monitoring and controlling growth. Since automatic weighing systems in broiler farms give less accurate readings for the weight of the birds towards the end of the growing period (Chedad *et al*., 2003), prediction techniques may help to give more accurate estimates of slaughter weight at the processing plant.

Prediction of growth patterns in animals has been studied for many years (Gompertz, 1825; Richards, 1959; Parks, 1965 and Fitzhugh, 1976 as cited by Parks, 1982). Research has continued on different species, comparing and contrasting different methods. Mignon-Grasteau and Beaumont (2000) investigated the main mathematical functions used to describe the growth curve for poultry species. In addition to three non-linear growth models (Richards, Gompertz and logistic), the spline linear regression model using body weight measurements from an unselected, random-bred chicken population was analysed by Aggrey (2002). It was concluded that growth parameters predicted with different models with fixed inflection points were not directly comparable. The relationship between the growth trajectory and carcase composition measurements was also investigated. Carcase yields among the different growth trajectories were similar within sexes. Aggrey (2004) used the spline linear regression to model pre-asymptotic body weight measurements from an unselected random-bred chicken population fed on two diets with different protein contents. He demonstrated that the use of the spline regression and relative growth models would have practical application for commercial broilers and broiler breeders where growth is either truncated before the asymptote or controlled by dietary and other environmental factors to prevent excessive weight gain.

The empirical equations, such as those of Richards and Gompertz, perceived growth as either white (internal operation known) or black boxes (internal operation unknown) with output only (Parks, 1982). Growth models have also been used to determine optimum feeding regimen. Models were developed to show the relationship between the feed consumed by an animal and the gain in body weight. Aerts *et al.* (2003*a*, *b*) calculated daily feed supply on the basis of a model-based control algorithm with the

Correspondence to: Ö. Cangar, Catholic University of Leuven, Faculty of Bioscience Engineering, Biosystems Department, M3-BIORES, Kasteelpark, Arenberg, B-3001 Heverlee, Belgium. Tel.: +32 (0)16321435. E-mail: ozlem.cangar@student.kuleuven.be

Accepted for publication 20th February 2006.

ISSN 0007–1668(print)/ISSN 1466–1799 (online)06/030330–6 © 2006 British Poultry Science Ltd
DOI: 10.1080/00071660600741735

aim of following a previously defined target growth trajectory as closely as possible. For the modelling of the dynamic growth response of broiler chickens to the control input (feed supply), online parameter estimation was used. The developed control algorithm was able to grow the birds according to different target trajectories ranging from restricted to compensatory growth.

Because the growth response of broilers to cumulative food intake was a non-linear process, Van Buggenhout *et al*. (2004) investigated the validity of the assumption of linearity in an adaptive modelling approach in terms of prediction accuracy. The dynamic growth response of the broiler chickens was modelled and predicted using a time-variant parameter estimation procedure. A comparison was made with the prediction accuracy of the recursive linear modelling approach and it was concluded that for growth control purposes the dynamic growth response of broiler chickens is slightly better modelled assuming non-linear dynamics in a short time window.

The objective of the present paper was to analyse different growth models and data fitting procedures to determine the best model for the end-weight prediction of broiler chickens prior to slaughter.

MATERIALS AND METHODS

Experimental data

For the purpose of this study, two data-sets were used. The first set was used for model development and the other was analysed for validation purposes.

Data-set 1

The first data-set contained two experiments where the daily weight response of broiler chickens during their growth to 42 d of age and their respective daily feed intake during this period were measured. In experiment 1, three different treatments over the growth period were applied, namely: *ad libitum* feed intake; 80% of *ad libitum* intake from d 4 until d 7; 80% of *ad libitum* intake from d 4 until d 11. There were two replicates per strain and per sex of each treatment, resulting in 24 different growth periods. Each experiment was conducted with 1200 Ross 508 and 1200 Hybro G chicks, adding up to a total of 2400 animals from both sexes. The birds were kept at 100 per pen. Each of the 24 pens contained an automatic weighing system (F47, Fancom BV, Panningen, the Netherlands) for measuring the average weight of the group of animals every 24 h. The daily amount of feed provided to the restricted groups was calculated on the basis of mean daily *ad libitum* feed intake during the previous day for chickens of the same strain and sex. More detail is given by Aerts *et al*. (2003*a*). In experiment 2, three different treatments were applied: *ad libitum* feed intake; *ad libitum* feed intake with a low energy diet; *ad libitum* feed intake with a sodium chloride deficient diet. For every treatment, two strains and both sexes were used. Each treatment per strain and per sex had two replicates. The experiments were conducted with exactly the same number of birds, using the same experimental design as experiment 1.

Data-set 2

All the experiments were conducted with mixed sex Ross 308 broilers. The birds were kept in two compartments on chopped straw with 2900 birds per compartment. Each compartment was equipped with 4 weighing platforms connected to a weighing computer (F47, Fancom BV). In compartment 1, birds were fed *ad libitum* while, in compartment 2, restricted growth curves were applied with end weights of 1945 and 2100 g (at d 42). A random sample of 200 birds was weighed every week in order to check the accuracy of the weighing system and to screen the variation within the group of birds. More information about data-set 2 can be obtained in Aerts *et al*. (2003*b*).

In order to test an alternative approach for end-weight prediction of broiler chickens, the mean weight at d 42 was predicted using different methods. The data from the weighing platforms for at most 15 d were used to calculate the end weight on certain days in the future. These predictions were then compared with the results from manual weighing. The prediction accuracies of the different approaches were evaluated and the best model was chosen according to statistical criteria (see below).

Prediction methods

Two different prediction methods were used and evaluated, namely, input–output and single output models. The input–output models included linear and non-linear recursive modelling with a time-varying model structure, whereas the output models consisted only of empirical growth equations and several alternative curve fitting techniques.

Input-output models

In the prediction of end weight, recursive modelling approaches with linear and non-linear time-varying model structures were used. The growth

response of the broiler chickens and the end weight (system output) were modelled using cumulative feed intake as the process input. In these recursive methods, the end weight was calculated up to 5 d ahead. In the model, the parameters were estimated in a recursive way by using a moving rectangular window approach. For both linear and non-linear approaches, daily averages of weight and food intake were used. The following linear (1) and non-linear regression equations (2) were used to describe the weight trajectory and thus predict the end weight of the chickens:

$$\text{Linear recursive}: \quad W(t) = \theta_{1t} + \theta_{2t} * CF(t) \qquad (1)$$

$$\text{Non-linear recursive}: \quad W(t) = W_0 + (A - W_0)(1 - \mathrm{e}^{-CF(t)/f}) \qquad (2)$$

where $W(t)$ is the measured average weight (g) at time t; $CF(t)$ is the measured cumulative food intake (g) at time t (d); θ_{1t} and θ_{2t} (kg/kg) are the time-variant model parameters estimated at time t; A is the mature weight (g); W_0 is the weight at time t_0 (g); and f is the amount of food consumption needed to increase the weight to 63·3% of $(A-W_0)$ (g). The model parameters of the two equations above were estimated in a recursive way (every day) by using a rectangular window approach, as described by Young (1984). A brief description of this approach is that, at every discrete time instant t, the parameters θ_{1t} and θ_{2t} are estimated from information on cumulative feed and weight measured during a time window of past samples. Window size is the number of past samples used in the approach. For all models, daily averages of weight and feed intake were used.

Single output models–curve fitting

End-weight prediction was carried out by fitting several curves to the growth trajectory of broiler chickens. Except for the condition of linear fitting (equation 3) all data from the past days were used to fit the best curve that would define the end weight at d 42 most accurately, 5 d ahead. This means that all 37 d of weight information was used to predict the weight at d 42. In linear fitting, 3 to 7 d of past information was used to fit a line that would predict 1 to 4 d ahead to d 42:

$$\text{Linear}: \quad W(t) = a * bt \qquad (3)$$

$$\text{Second-order polynomial}: \quad W(t) = at^2 + bt + c \qquad (4)$$

$$\text{Third-order polynomial}: \quad W(t) = at^3 + bt^2 + ct + d \qquad (5)$$

$$\text{Gaussian}: \quad W(t) = a * \mathrm{e}^{-[(t-b)/c]^2} \qquad (6)$$

$$\text{Gompertz}: \quad W(t) = a^* \mathrm{e}^{-\mathrm{e}^{[(t-b)/c]}} \qquad (7)$$

where $W(t)$ is the weight of the broiler chickens at time t(g); t is the time in d; a, b and c are the coefficients of the equations.

Statistical analysis

The goodness of fit of the end-weight prediction techniques was calculated by using mean relative prediction error (MRPE) defined as

$$\mathrm{MRPE}_{42} = \sqrt{\left[\frac{w_{42} - \hat{w}_{42}}{w_{42}}\right]^2} * 100$$

where MRPE_{42} is a percentage; w_{42} is the weight measured with the automatic weighing systems at d 42 (g); $\hat{w}_{42}$ is the predicted weight at d 42 (g). To find out how widely the weight values were dispersed from the mean value for the bird population on d 42, standard deviation was calculated for 200 manual weighing measurements for each treatment of the validation dataset. The significance of differences in mean relative prediction errors obtained by different models were tested using the analysis of variance method.

RESULTS

Model development

The recursive input–output method, using the information in its window size, predicted the weight 5 d ahead by taking into account the feed intake (input) between d 37 and d 42. Since feed intake information was not available for these future days, it was also predicted by means of a third-order polynomial fit to the feed intake data of the previous 37 d.

To find the optimum window size and prediction horizon that would provide the least prediction error in a recursive method, several graphs were drawn with MRPE of the end weight as a function of window size and prediction horizon. When the MRPE's of linear and non-linear recursive methods were compared, it was seen that a non-linear method with an average prediction error of 2·8% gave more accurate results than the linear method (4·0%) in end-weight prediction. It can be seen in Figure 1 that, for both cases, MRPE increased with increasing prediction horizon but, interestingly, was minimal around d 15. This shows the challenge of using the recursive method to follow the growth for more than 15 d. In fact large window sizes did not necessarily decrease the end-weight prediction error. A window size of 15 d and prediction horizon of 5 d were chosen for further evaluation of MRPE in the recursive method.

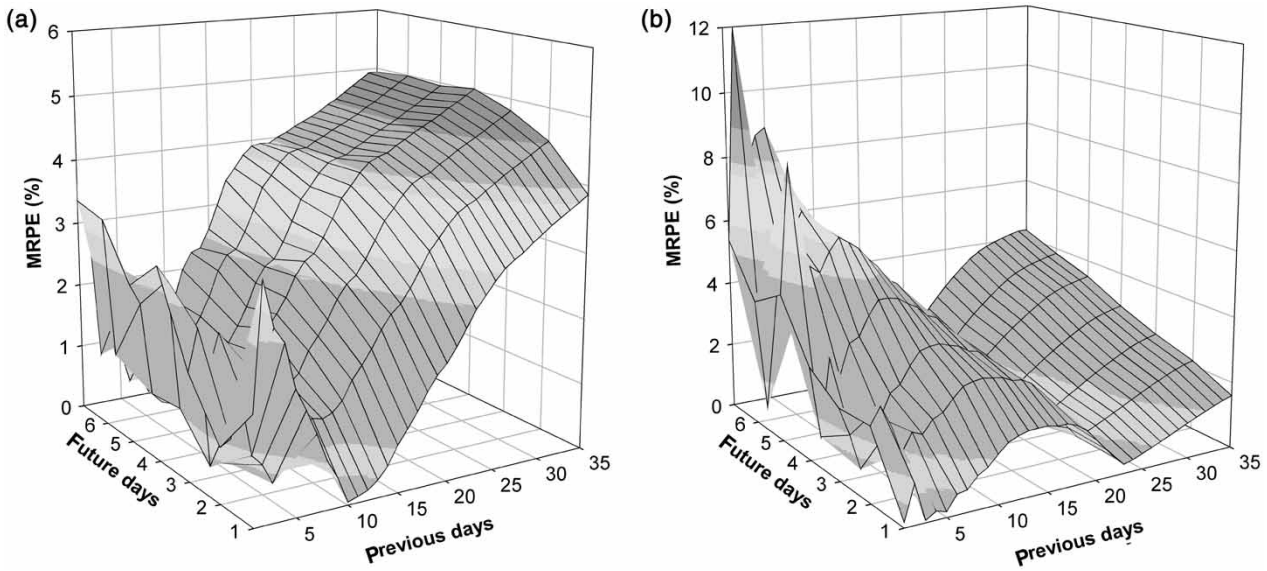


Figure 1. *Examples of mean relative prediction error (MRPE) for the linear (a) and non-linear model (b) as a function of window size (previous) and prediction horizon (future days).*

Table 1. *Mean relative prediction errors (% and g) for the end weight of broiler chickens 5 d ahead, using the linear and non-linear recursive methods, together with linear, polynomial and Gaussian curve fitting and the Gompertz empirical model (using data-set 1)*

	MRPE (%) of data-set 1		
METHOD	Experiment 1	Experiment 2	Average
Linear recursive (15, 5)[1]	2·19	5·80	4·00
Non-linear recursive (15, 5)	2·97	2·66	2·82
Linear (32:37)[2]	3·60	2·76	3·18
Second-order polynomial	3·20	5·70	4·45
Third-order polynomial	3·26	4·23	3·75
Gaussian	10·11	4·84	7·48
Gompertz	3·76	2·78	3·27

[1] Non-linear/linear recursive (n_1, n_2) stands for non-linear recursive method with window size of n_1 and prediction horizon of n_2 d.
[2] Linear ($n_3 : n_4$) stands for the linear curve fitting taking into account the information of the d n_3 to n_4.

All the predictions, except linear fitting, took account of the previous 37 d to predict the last 5 d. Linear fitting, on the other hand, used only the data from d 32 to d 37 to predict the last 5 d. The mean relative errors of end-weight prediction in two experiments using different methods are given in Table 1. Analysis of variance of the predicted means of the different methods were significantly different from each other ($F = 3{\cdot}65$, $P = 0{\cdot}004$).

Model validation

Since the least error in end-weight prediction was obtained from the non-linear recursive and linear fitting methods, these were used for further evaluation of the best method on data-set 2. The results can be seen in Figure 2. In Tables 2 and 3, mean relative prediction errors obtained using these two best methods are given. Linear fitting, on average, provided the best results for prediction horizons of 4, 3 and 2 d. Therefore, in the validation, linear fitting models for different numbers of past and future days were evaluated. Figure 3 shows the mean relative prediction errors for linear fittings as a function of past and future days.

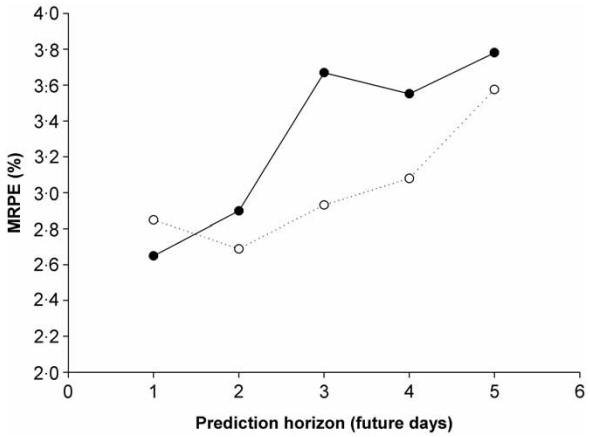


Figure 2. *Mean relative prediction errors (MRPE) vs prediction horizon in non-linear recursive (window size of 15 d and prediction horizon of 5 d, –●–) and linear fitting models (window size of 5 d and prediction horizon of 5 d, · · ·○· · ·).*

Table 2. *Mean relative prediction errors (%) for the end weight of broiler chickens using the non-linear recursive method with window size of 15 d (validation with data-set 2)*

		MRPE (%) of data-set 2				
		Prediction horizon (d)				
Data-set 2	End weight (g)	5	4	3	2	1
Repetition 1-rest[1]	1964	2·26	2·13	2·04	1·69	1·27
Repetition 1-ad lib	2219	0·25	0·06	0·27	0·21	0·08
Repetition 2-rest	1826	3·38	3·76	3·81	4·27	4·45
Repetition 2-ad lib	1793	5·34	5·82	7·22	8·25	9·14
Repetition 3-rest	2076	3·26	3·20	3·06	2·70	2·27
Repetition 3-ad lib	2150	4·53	4·37	4·03	3·45	2·83
Repetition 4-rest	1861	3·56	3·07	2·07	0·89	0·20
Repetition 4-ad lib	1897	6·06	4·37	2·72	1·50	0·67
Repetition 5-rest	1923	3·82	3·25	5·46	1·50	0·61
Repetition 5-ad lib	1797	1·65	1·59	3·66	0·15	0·60
Repetition 6-rest	1721	12·32	11·40	10·69	10·54	10·72
Repetition 6-ad lib	1907	6·52	6·68	6·32	5·43	4·25

[1] The abbreviations rest and ad lib stand for different repetitions of data-set 2 with either *ad libitum* (ad lib) or with restricted (rest) feeding regimes.

Table 3. *Mean relative prediction errors (%) for the end weight of broiler chickens using the linear curve fitting method with window size of 5 d (validation with data-set 2)*

		MRPE (%) of data-set 2				
		Prediction horizon (future d)				
Data-set 2	End weight (g)	5	4	3	2	1
Repetition 1-rest[1]	1964	0·55	0·20	0·01	0·04	0·03
Repetition 1-ad lib	2219	1·29	0·44	0·01	0·01	0·01
Repetition 2-rest	1826	0·62	3·23	4·04	5·65	5·62
Repetition 2-ad lib	1793	3·65	6·99	10·18	11·15	11·57
Repetition 3-rest	2076	5·39	3·28	1·94	1·29	0·94
Repetition 3-ad lib	2150	3·34	2·72	2·09	1·73	1·67
Repetition 4-rest	1861	5·66	3·69	0·32	2·14	3·01
Repetition 4-ad lib	1897	3·68	3·08	1·09	0·02	0·72
Repetition 5-rest	1923	5·35	2·27	3·54	2·22	2·32
Repetition 5-ad lib	1797	2·05	1·41	4·43	1·36	2·19
Repetition 6-rest	1721	7·85	8·47	8·74	9·82	10·87
Repetition 6-ad lib	1907	10·63	7·34	4·65	2·23	0·92

[1] The abbreviations rest and ad lib signify different treatments with either *ad libitum* (ad lib) or with restricted (rest) feeding regimes.

DISCUSSION

Several methods were analysed and compared for prediction of the mean slaughter weight of broiler chickens some days in advance. Predictions were carried out using two methods, namely, input–output and single output models. Input–output models, in contrast to the single output model, included feed intake as the input to the common output, the weight response. Linear and non-linear recursive methods were applied for the first and simple curve fitting for the second method. As expected, the non-linear recursive model was more accurate in predicting the end weight. The empirical models such as third-order polynomial, Gaussian and Gompertz curves, despite their goodness of fit, did not give such good results as non-linear recursive or the simple linear fit.

In further evaluation of the best two models, non-linear recursive and linear fitting, it was observed that the latter provided more accurate results for prediction of end weight, 4, 3 and 2 d ahead. When the number of future days was increased, mean relative prediction error also increased. Conversely, as the number of past days increased, error decreased. It is thus better to use the information from more past days to fit a linear line to the growth curve. Even though there is a lack of information in the literature for end-weight prediction of broiler chickens, it can be concluded that the simple linear growth curve fitting method gives the best accuracy. Error is minimised to an average of 0·14% when 4 d of past information is used to fit a line to predict the end weight, one day ahead.

As shown in the literature (Rogers *et al.*, 1987), linear regression applied to short intervals of the growth curve is an accurate means of interpreting growth rates. Growth rates during the 5 d prior to processing are therefore almost

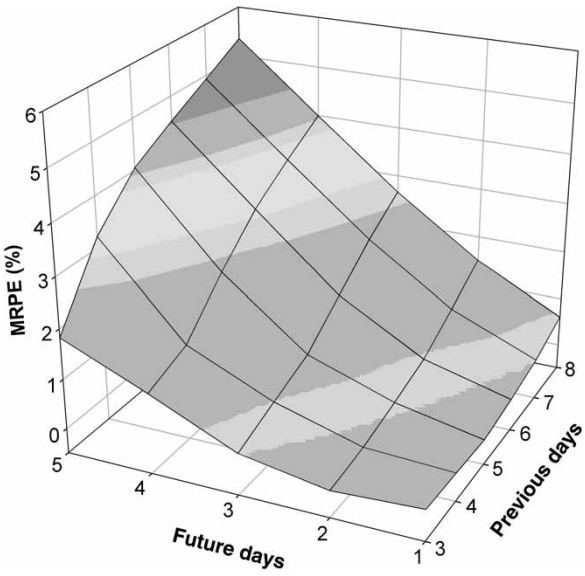


Figure 3. *Mean relative prediction errors (MRPE, %) for linear fittings as a function of previous and future days.*

constant and the end weight can most accurately be predicted by means of a linear extrapolation of the growth rates in the previous 5 d. Other methods involving growth curves etc. are not as accurate because of short-term changes that might occur in those 5 d, but accuracy can be improved with these methods by the use of weights over a longer period of time.

Finally, when transportation weight losses are considered, slaughter weight should be decreased by 0·2 to 0·5% for every hour of transportation to the processing plant (Elrom, 2000).

REFERENCES

Aerts, J.M., Lippens, M., De Groote, G., Buyse, J., Decuypere, E., Vranken, E. & Berckmans, D. (2003*a*) Recursive prediction of broiler growth response to feed intake by using a time-variant parameter estimation method. *Poultry Science*, **82**(1): 40–49.

Aerts, J.M., Van Buggenhout, S., Vranken, E., Lippens, M., Buyse, J., Decuypere, E. & Berckmans, D. (2003*b*) Active control of the growth trajectory of broiler chickens based on online animal responses. *Poultry Science*, **82**(12): 1853–1862.

Aggrey, S.E. (2002) Comparison of three nonlinear and spline regression models for describing chicken growth curves. *Poultry Science*, **81**(12): 1782–1788.

Aggrey, S.E. (2004) Modelling the effect of nutritional status on pre-asymptotic and relative growth rates in a random-bred chicken population. *Journal of Animal Breeding and Genetics*, **121**(4): 260–268.

Chedad, A., Aerts, J.M., Vranken, E., Lippens, M., Zoons, J. & Berckmans, D. (2003) Do heavy broiler chickens visit automatic weighing systems less than lighter birds? *British Poultry Science*, **44**(5): 663–668.

Elrom, K. (2000) Handling and transportation of broilers welfare, stress, fear and meat quality, Part V: Transport to the slaughterhouse. *Israel Journal of Veterinary Medicine*, **56**(1): 1–3.

Fitzhugh, H.A. (1976) Analysis of growth-curves and strategies for altering their shape. *Journal of Animal Science*, **42**(4): 1036–1051.

Mignon-Grasteau, S. & Beaumont, C. (2000) Growth curves in birds. *Productions Animales*, **13**(5): 337–348.

Parks, J.R. (1982) *A Theory of Feeding and Growth of Animals* (Berlin, Springer-Verlag).

Rogers, S.R., Pesti, G.M. & Marks, H.L. (1987) Comparison of 3 nonlinear-regression models for describing broiler growth-curves. *Growth*, **51**(2): 229–239.

Van Buggenhout, S., Aerts, J.M., Vranken, E. & Berckmans, D. (2004) Non-linear modelling of broiler growth using a time-variant parameter estimation method. *Transactions of the American Society of Agricultural Engineers*, **47**(5): 1757–1764.

Young, P.C. (1984) *Recursive Estimation and Time Series Analysis: An Introduction* (Berlin, Springer-Verlag).

Copyright of British Poultry Science is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.